

# NAMAN MANOCHA

AI ENGINEER • GENAI • AGENTIC SYSTEMS — LLMs, RAG & Applied ML

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## SUMMARY

AI/ML engineer with experience across classical ML, multi-agent LLM orchestration, and grounded RAG systems. Independently developed FundersAI, a research platform for Indian stocks and mutual funds, and evidence-first AI product work across Python, FastAPI, Next.js, PostgreSQL, Docker, and Kubernetes.

## TECHNICAL SKILLS

ML & AI: LLMs, LangGraph, Google ADK, RAG, Hybrid Search, Prompt Engineering, Hugging Face, Ollama, pgvector, ChromaDB, K-Means Clustering, Logistic Regression, Langfuse, MLflow, scikit-learn

Backend & Cloud: Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Redis, Docker, Kubernetes, GitHub Actions, Google Cloud Run, Vercel, Prefect, Git

Web & Data: Next.js, React, TypeScript, ETL Pipelines, Web Scraping, PDF Parsing, Data Validation, Data Visualization

## EXPERIENCE

### Technology Manager — PayGain Multiservices Pvt. Ltd.

Feb 2025 – Present

Kolkata, India

- Scoping and leading development of a company-wide mobile app as sole engineer, covering client/employee onboarding, document access, product activation, employee training, and payments — mobile counterpart to PayGain's core platform.
- Designed and deployed a lead-capture automation (Make) triggered on Google Form submissions, eliminating manual lead logging and improving sales outreach.
- Drove 30+ client conversions by building and managing the company website and digital discovery flows.
- Simplified complex investment/insurance products for 300+ seminar attendees via 20+ technical presentations and collateral — translating business requirements into deployed, client-facing systems.

## INDEPENDENT PROJECTS

### FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

[fundersai.co.in](https://fundersai.co.in)

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2, GitHub Actions

Solo-built, Apr–Aug 2026 — ~137K source LOC across 826 files, 434 commits, 167 test files — submitted to OpenAI Build Week | Live: [www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)

- **Citation-Grounded RAG:** built a seven-stage LangGraph workflow with retrieval grading, query rewriting, citation-aware synthesis, lexical reranking, and explicit abstention when evidence is insufficient.
- **Explainable ML:** engineered a fund-similarity engine using cosine similarity and k-means clustering over 13 standardized features, plus a logistic-regression triage model (MLflow) for parser-review prioritization.
- **Multi-Agent Orchestration:** designed a custom multi-agent pipeline (Planner → Quant → Research → Synthesis agents) with shared state and real-time streamed status updates, where a Groq-hosted LLM router agent classifies query intent and directs execution across stock, mutual-fund, comparison, and market-news paths.
- **Data-Quality Debugging:** diagnosed and fixed scheme-name normalization defects, then added safeguards to preserve correct mutual-fund grouping and coverage.
- **Resilient Data Engineering:** automated official disclosure collection and normalization with source-specific parsers, data-readiness checks, and manual promotion gates.
- **Cloud Storage & Billing:** built an S3-compatible storage layer (Cloudflare R2 via boto3/AWS SigV4, Python + TypeScript) for raw documents and cold archival, and shipped Razorpay subscription billing (webhook-verified, Free/Pro/Ultra) with Supabase Auth row-level security.

*Python, FastAPI, LangGraph, Docker, Kubernetes (K3s), AWS EC2*

- **Decoupled Architecture:** extracted report synthesis into a standalone FastAPI + LangGraph service, independently deployable from the main monolith so LLM-heavy generation scales without impacting the core app.
- **Kubernetes Service Operations:** provisioned and administered an AWS EC2 instance running K3s, hosting the report-synthesis service with two replicas behind a streaming API for real-time fund-comparison report generation.
- **Infrastructure & Secrets:** configured EC2 security groups/networking and managed credentials (Supabase, OpenAI) via Kubernetes Secrets rather than embedding them in the container image.

### CareFlow Intelligence — Agentic RAG & Fine-Tuning for Clinical Research

*Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | Solo-built, Aug 2026 — ~7,300 LOC — non-clinical, synthetic-data-only (MITRE Synthea)*

- **Agentic Citation-Grounded RAG:** built a bounded-tool document agent with query reformulation and mandatory citation validation against PostgreSQL full-text indexes, backed by automated source ingestion.
- **RAFT Fine-Tuning:** designed a training-data pipeline with reference evidence, hard distractors, and abstention cases; fine-tuned Qwen2.5-1.5B via 4-bit QLoRA (rank 16, alpha 16), then quantized and served it locally through Ollama.
- **Local Model Serving:** quantized the fine-tuned model to GGUF and deployed it behind Ollama for zero-cost local inference, with an OpenRouter (Nemotron 3 Ultra) cloud fallback for the base agent.
- **Human-in-the-Loop Data Agent:** built a second agent for synthetic clinical data ingestion (10 patients, 479 encounters, 300 conditions, 5,368 observations, 296 medications across 18 FHIR R4 tables) with mandatory human-approval gates around autonomous data operations.
- **Full-Stack Delivery:** async FastAPI + SQLAlchemy + Alembic backend and Next.js 15 frontend, Dockerized end-to-end for local and cloud deployment.

### TalentOS — Autonomous Opportunity Intelligence Platform (All Things Agentic Hackathon)

*Python, Google ADK, LangGraph, Next.js, Firestore, Google Cloud Run | Solo-built, Aug 2026 — ~21K source LOC across 116 files, 70 commits, 26 test files | Live: [all-things-agentic--allthingsage...](#) | Repo: [github.com/Naman6019/all-things-agentic](#)*

- **Dual-Pipeline Agent System:** built Careers and Studio workflows with LangGraph StateGraphs and Google ADK agents on Gemini, aggregating public ATS and freelance opportunity sources into reviewable outputs.
- **Cost-Disciplined Filtering & Evaluation:** applied deterministic pre-filtering before model evaluation, then used structured evaluator and drafter steps to prepare tailored application materials with persistent run state.
- **Guardrails & Observability:** shipped human-in-the-loop send gates, platform-compliance checks, Langfuse tracing, and a hermetic offline test suite for agent behavior.
- **Full-Stack Dashboard:** built a Next.js 15/React 19 dashboard with separate jobs and freelance views, printable tailored resumes, and an interactive pitch editor, deployed on Cloud Run.

### Question Answering on SQuAD — Model Fine-Tuning & Benchmarking

*Python, PyTorch, scikit-learn, BERT, BiDAF, DistilBERT*

- Compared BERT, BiDAF, and DistilBERT for extractive QA; fine-tuned each and built a standardized evaluation pipeline (accuracy, precision, recall, F1); published as a research paper.

### GitResume — Autonomous Resume & Portfolio Intelligence Agent

*Python, Typer, Rich, Ollama, python-docx | ~4K LOC across 23 files, 27 commits | Repo: [github.com/Naman6019/git-resume-agent](#)*

Self-built a multi-agent pipeline (Inspector → SchemaDiscoverer → Synthesizer → GroundingVerifier) that scans git history/AST, synthesizes quantified resume bullets via LLM, and adversarially verifies every metric against commit diffs before compiling to Word/PDF with native hyperlinks — published to PyPI as git-resume-agent; runs on every commit via a git hook.

## EDUCATION & ACHIEVEMENTS

### B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati

*June 2021 – Sept 2025*

CGPA: 8.2/10

- Published research paper: "Interactive Question Answering Model using Natural Language Processing" — International Journal of Modern Developments in Engineering and Science.
- Certifications: Azure Administrator Associate; AWS Certified Cloud Practitioner (Foundational).